1. Data Generation Approach

The test dataset (test_set.json) was synthetically generated using OpenAI's gpt-4o-mini model through a dedicated TestDatasetGenerator script. To ensure high quality, diversity, and soundness, specific prompting strategies were applied for each category:

- **Basic Queries:** The prompt enforced strict boundaries. For instance, the relevant category was constrained to topics explicitly covered in the provided Machine Learning textbook (e.g., Decision Trees, KNN, Bias-Variance tradeoff) to ensure grounded RAG testing.
- **Advanced Queries:** For hybrid queries, the generator was instructed to combine a valid ML topic and an irrelevant/obnoxious question in a single string. For multi_turn scenarios, the LLM was formatted to output a list of lists, simulating sequential context retention (e.g., asking about a concept, then asking a follow-up pronoun question) and behavior recovery (e.g., a rude remark followed by a valid question).

2. Final Metric & Performance

The evaluation was conducted using an LLM-as-a-Judge methodology (GPT-4o-mini), performing binary classification based on specific criteria for each category. The overall system achieved an excellent **96.00% Accuracy**.

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FINAL EVALUATION REPORT
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Category: obnoxious       | Accuracy: 100.00%
Category: irrelevant      | Accuracy: 100.00%
Category: relevant        | Accuracy: 100.00%
Category: small_talk      | Accuracy: 100.00%
Category: hybrid          | Accuracy: 100.00%
Category: multi_turn      | Accuracy: 71.43%
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OVERALL SYSTEM ACCURACY: 96.00%
==============================
```

3. Failure Modes and Agent Weaknesses

Despite the high overall accuracy, the evaluation highlighted a few inherent weaknesses and edge cases in the multi-agent design:

- **Strict Anti-Hallucination vs. Judge Expectations (The "I don't know" Penalty):** In Multi-turn Prompt 1, the user asked about the disadvantages of a Decision

Tree. The RAG system correctly recognized that the provided textbook chunk did not specify the disadvantages and replied, *"I couldn't find relevant material to answer that."* While this is the safest and most desired behavior for a RAG system (zero hallucination), the rigid LLM Judge penalized it with a score of 0 because it expected a factual ML answer. This reveals a gap between strict factual grounding and conversational expectations.

- **Context Contamination in Domain Shifts:** In Multi-turn Prompt 4, Turn 1 was an irrelevant ice cream question, which the agent correctly handled via the Small_Talk or Topic_Check fallback. However, in Turn 2, when the user shifted to a valid ML question ("advantages of Ensemble Methods"), the agent failed and triggered the topic_check refusal. This indicates a weakness in the Controller's history management: when a previous turn introduces heavy out-of-domain context, the Query_Agent can become overly sensitive and misclassify the subsequent valid queries as irrelevant.
- **Rigid Pipeline Bottlenecks:** The system operates on a linear gatekeeping model (Obnoxious -> Topic Check -> Retrieve -> Relevance -> Answer). A weakness of this design is that if the topic_check prompt is not perfectly balanced, complex queries can be prematurely blocked.